

UNIVERSITY PARK, PA, USA

WMO: 725128

Lat: 40.850N

Lon: 77.850W

Elev: 1239

StdP: 14.05

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 54739

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.7	(c) 10.3	(d) -6.1	(e) 4.1	(f) 10.3	(g) -1.2	(h) 5.4	(i) 12.6	(j) 26.6	(k) 35.3	(l) 24.3	(m) 30.9	(n) 8.4	(o) 270	(p) 0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.3	(c) 88.5	(d) 71.0	(e) 85.9	(f) 70.1	(g) 82.5	(h) 68.5	(i) 74.2	(j) 82.6	(k) 72.7	(l) 80.9	(m) 71.4	(n) 79.3	(o) 8.8	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.2	124.8	77.5	70.4	117.0	76.1	69.5	113.6	75.6	38.6	82.5	37.1	81.5	36.0	80.2	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.9	(b) 19.0	(c) 16.8	(e) 2.4	(f) 91.6	(g) 3.6	(h) 3.1	(i) -0.2	(j) 93.8	(k) -2.4	(l) 95.7	(m) -4.4	(n) 97.4	(o) -7.1	(p) 99.7		
(a) 22.9	(b) 19.0	(c) 16.8	(e) 1.0	(f) 76.4	(g) 3.6	(h) 1.7	(i) -1.6	(j) 77.6	(k) -3.7	(l) 78.7	(m) -5.7	(n) 79.6	(o) -8.4	(p) 80.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	27.6	29.6	38.1	50.3	59.3	67.7	71.4	70.8	63.4	52.0	42.4	31.1		
	DBStd	17.62	10.91	9.10	9.57	9.05	7.64	6.22	5.12	5.17	6.33	7.93	8.79	8.90		
	HDD50	2696	699	573	388	105	11	0	0	0	1	71	258	590		
	HDD65	5948	1160	992	834	450	213	44	6	10	103	408	678	1050		
	CDD50	2848	4	1	20	113	298	530	665	646	402	134	30	5		
	CDD65	625	0	0	1	8	35	124	206	191	54	6	0	0		
	CDH74	4835	0	0	8	123	321	951	1633	1361	391	45	2	0		
	CDH80	1328	0	0	1	20	67	245	500	401	86	8	0	0		
	WSAvg	6.3	7.9	8.1	7.9	7.5	5.8	5.0	4.5	4.0	4.6	5.9	6.7	7.9		
	PrecAvg	38.4	2.7	2.5	3.2	3.0	3.5	3.9	3.5	3.5	3.6	2.8	3.3	2.7		
(15) Precipitation	PrecMax	59.3	7.6	5.7	6.8	8.5	6.5	12.8	7.3	9.1	11.0	7.5	7.4	5.9		
	PrecMin	28.0	0.5	0.4	0.9	0.4	0.9	0.3	0.8	0.4	0.9	0.1	0.7	0.6		
	PrecStd	7.4	1.6	1.2	1.3	1.7	1.5	2.2	1.5	1.9	2.1	1.8	1.6	1.3		
	PrecStd	7.4	1.6	1.2	1.3	1.7	1.5	2.2	1.5	1.9	2.1	1.8	1.6	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.0	59.2	74.6	82.3	86.3	89.7	92.1	91.2	87.1	79.0	70.4	63.6		
		MCWB	53.3	51.0	58.0	61.7	68.5	70.9	72.7	71.3	68.0	66.2	58.9	54.7		
	2%	DB	55.0	52.0	65.5	77.4	81.9	86.1	89.7	88.1	81.9	73.1	64.1	54.9		
		MCWB	50.5	43.3	52.5	58.6	65.2	70.4	72.1	71.2	66.4	59.7	55.7	48.0		
	5%	DB	48.2	46.3	59.2	71.9	78.6	82.4	85.7	84.1	78.8	69.9	61.0	48.0		
		MCWB	44.4	39.8	48.7	56.4	63.5	68.9	70.5	70.7	65.4	58.4	52.9	43.3		
	10%	DB	41.9	43.0	54.2	66.0	73.3	80.5	82.2	81.2	74.9	65.7	56.8	43.2		
		MCWB	38.1	37.8	46.0	52.9	60.7	67.6	68.9	68.4	64.0	58.1	50.3	38.7		
	0.4%	WB	56.7	52.1	59.2	65.2	70.7	75.0	76.7	76.2	72.2	67.8	62.2	57.2		
		MCDB	59.5	58.3	70.5	78.0	83.0	83.0	86.5	82.8	80.1	77.4	66.8	61.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	51.2	45.1	55.1	60.8	68.2	72.8	74.5	73.8	70.0	64.0	57.6	49.5		
		MCDB	54.6	49.2	63.0	73.1	78.3	81.4	83.3	81.8	76.4	68.7	63.7	54.1		
	5%	WB	44.1	41.1	50.5	58.3	66.1	71.2	72.8	72.4	68.1	60.9	54.4	44.0		
		MCDB	47.9	46.0	58.8	68.1	74.4	79.6	81.1	80.1	73.9	67.8	59.6	47.0		
	10%	WB	38.5	37.7	46.2	55.0	63.8	69.5	71.4	71.0	66.6	57.9	51.0	39.2		
		MCDB	41.8	42.7	52.3	64.4	70.3	76.6	79.4	78.1	72.5	64.1	55.8	43.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	14.0	16.5	19.7	19.7	18.6	19.3	18.2	18.5	17.1	14.4	11.8		
		MCDBR	18.7	20.2	23.9	28.4	24.8	22.3	22.5	22.2	24.1	24.5	20.3	18.3		
		MCWBR	15.7	15.5	14.7	14.9	11.9	9.8	8.6	9.0	11.4	13.3	14.3	15.3		
		MCDBR	17.3	18.4	21.6	24.3	20.7	18.6	18.2	18.0	17.3	21.2	17.9	16.6		
	5% WB	MCWBR	15.7	15.3	15.4	14.5	10.8	9.4	8.7	8.4	9.8	13.1	14.8	15.2		
		MCWBR	15.7	15.3	15.4	14.5	10.8	9.4	8.7	8.4	9.8	13.1	14.8	15.2		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.326	0.353	0.377	0.414	0.472	0.483	0.475	0.388	0.370	0.326	0.305		
	taud		2.397	2.333	2.352	2.327	2.246	2.107	2.092	2.099	2.331	2.373	2.476	2.466		
	Ebn at Noon		267	279	284	284	274	258	253	251	269	260	260	261		
	Edh at Noon		26	32	36	39	43	50	50	49	36	31	24	23		
(44) All-Sky Solar Radiation	RadAvg		516	794	1095	1439	1626	1786	1809	1605	1305	890	607	423		
	RadStd		40	95	94	144	170	153	121	92	166	100	71	53		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+198	N/A

Nomenclature: See separate page